Tile graph communication inside the direct U-Net

V5-D prototype | One tile = one node; graph context conditions the raster forecast

1 Shared tile encoder

Terrain + observed state/history
Forcing, times and requested lead
Optional hydraulic halo
Encode every context tile

2 Node features

Pooled 20 × 20 spatial features
Four directional boundary summaries
Tile forcing/time/lead conditioning
Linear node projection

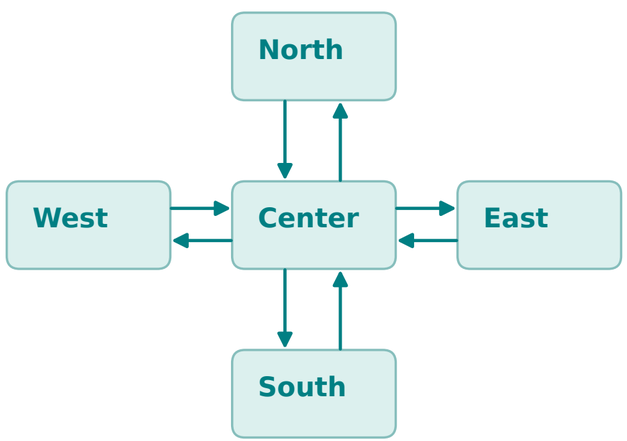
3 Message passing

Learn messages on tile edges
Mean incoming messages
Residual node update
Repeat K spatial layers

4 U-Net bottleneck context

$c' = c + P(z^K - z^0)$
Concatenate with pooled raster features
Decoder + original skip features
Central depth, extent, HU/HV forecast

Shared-boundary tile adjacency



Graph sampling contract

Connected tiles share one event, origin and lead.
Expand context to cover the graph layers.
Use observed inputs; score designated centers.
Local communication does not cover the entire basin.

What the graph adds

Halo: nearby pixel-level hydraulic context.
Graph: learned tile-to-tile feature communication.
Zero-initialized projection starts with no added context.
HUC12 hierarchy remains a proposed extension.

Edges encode geometric contact. Learned messages are features, not conserved water fluxes.

Implemented local graph prototype; corrected-data forecasting benefit remains unproven.